Answers of training book

Functions of superficial fascia

-Stores water and fat.

- Reduces the rate of heat loss

- provides mechanical protection against traumatic blows.

- provides a framework for nerves and blood vessels to enter and to exit muscles

Functions of deep fascia

-allows free movement of muscles

- supports nerves, blood vessels and lymphatic vessels

- fills spaces between the muscles

Bones provide the following functions

- Protection for vital structures.
- Support for the body.
- The mechanical basis for movement.
- Storage for salts (e.g., calcium).
- A continuous supply of new blood cells (produced by the bone marrow within many bones).

Types of bones in the human body

1. long bones example humerus, ulna, radius, femur, tibia, fibula

2. short bones example carpal , tarsal bones

3. flat bones example scapula, skull , sternum

4. irregular bones example vertebrae , sacrum

5. sesamoid bones example patella

6- pneomatic bones example para nasal sinuses

The vertebral column consists of 33 number of vertebrae consists of

1. Cervical: …7…. number vertebrae
2. Thoracic: …12…. number vertebrae
3. Lumbar : …5…. number vertebrae
4. Sacrum: …5…. number vertebrae
5. Coccyx: …3-4…. number vertebrae

About joints complete the followings

- **Immoveable joints called fibrous joints**
- **Growing bones contains……1ry cartilagenous…..type of joints**
- **Freely moveable joints called synovial joints**
- **synovial membrane secretes synovial fluid**

About Types of Synovial Joint complete the followings

- **uniaxial hinge synovial joint permitted movement in one plane around a transverse axis**
- **biaxial condyloid synovial joint in which articular surfaces include an oval convex surface fitting into a concave surface**
- **multiaxial ball and socket synovial joint in which articular surfaces include a rounded head fitting into a cup shaped socket**

 plane synovial joint in which opposed surfaces of the bones are flat

About Movements at joints

 A bending movement that decreases the angle between two parts called flexion

 Moving away from the median plane of the body called abduction

 Rotation towards the axis of the body called internal rotation

What are the three types of muscles?

1. skeletal muscle example biceps
2. cardiac muscle example muscle of heart
3. smooth muscle example muscles of stomach

As regards the Cardiovascular System complete the followings

Right atrium contains …(deoxygenated)…..types of blood

Left ventricle contains …(oxygenated)…..types of blood

All arteries carry oxygenated blood with the exception of pulmonary artery

All veins carry deoxygenated blood except pulmonary vein

What Are The Differences Between Arteries And Veins?

1)Arteries are the carriers of red oxygenated blood directly from the heart, whereas veins carry de-oxygenated blood towards the heart.

2) Blood is under great pressure in arteries; hence it flows fast. In veins the blood is not under great pressure, hence it flows more slowly and smoothly.

3)Arteries have thick muscular walls. While veins have relatively thin muscular walls.

4) Arteries have no valves but veins have internal valves along their length to prevent back flow of blood.

As regards the respiratory System complete the followings

▯ Pharynx divided into three sections nasopharynx ,oropharynx ,laryngeopharynx

▯ Organ responsible for the production of voice is larynx

▯ Right lung: divided into ……3…….lobes

▯ The …left…. lung has a cardiac notch on its anterior border

As regards the digestive System complete the followings

▯ Organ extends from the pharynx to the stomach is oesophagus .

▯ Fixed part of small intestine is duodenum

▯ The most dilated part of the large intestine is cecum

▯ Pelvic colon is the other name of …sigmoid..colon

- The largest of the salivary glands is parotid gland
- The pancreas composed of 4 parts: head, neck, body, tail

The large intestine differs from the small intestine by the presence of the following main features

- Taeniae coli: these are 3 longitudinal muscular bands in the wall of the colon
- Sacculations (haustrations): these are sac-like dilatations in the wall of the large intestine especially in the colon.
- Appendices epiploicae: these are small projections of fat scattered over the surface of the colon.

As regards the urinary System complete the followings

- On the upper end of each kidney the supra renal gland rests
- Each kidney consists of millions of functional units called nephrons
- Male urethra divided into 3 parts: prostatic, membranous, spongy
- Female urethra is ……4….cm.

As regards the female genital System complete the followings

- Muscular tubes that extend from the left and right superior corners of the uterus to the edge of the ovaries called fallopian tubes
- The lower part of the uterus known as cervix

 Connects the uterus to the exterior of the body is vagina

 The inner lining of the uterus, known as the endometrium

As regards the male genital System complete the followings

 A sac-like organ made of skin and muscles that houses the testes is called scrotum

 A sperm storage area is epidydimis

 vas deferes carries sperm superiorly from the epididymis into the abdominal cavity to the ejaculatory duct

 The urethra in males is a common pathway for urine and semen

Main functions of the lymphatic system

 To collect and return interstitial fluid(surrounds the tissue cells), including plasma protein to the blood, and thus help maintain fluid balance (this means it collects liquid lost from the circulatory system- returns proteins to circulation)

 To absorb lipids from the intestine and transport them to the blood. (this means it transports fats to circulation)

 To defend the body against disease by producing lymphocytes and carries bacteria to the lymph nodes for destruction

As regards the Nervous System complete the followings

 Central nervous system (CNS), made up of brain and spinal cord

There are …12…….pairs of cranial nerves and ……31……. pairs of spinal nerves

The outermost layer of the meninges is the dura matter

The Cerebrospinal Fluid (CSF) circulates between

Arachnoid matter and pia matter of the meninges

General Embryology

As regards the human chromosomes complete the followings

A. Each cell normally contains……22….. Pairs of autosomes and ……1……pairs of sex chromosomes.

B. Chromosomes are divided into ……92……..number of chromatids

What are the differences between mitosis and Meiosis?

A. During mitosis one cell divides once to form two identical cells, During meiosis one cell Divides twice to form four daughter cells These four daughter cells only have half the number of chromosomes of the parent cell – they are haploid.

B. The major purpose of mitosis is for growth and to replace worn out cells, Meiosis produces our sex cells or gametes

As regards the Primary sex organs complete the followings

a) Primitive male germ cell is called spermatogonium While Primitive female germ cell is called oogonium

b) Supporting cells in the seminiferous tubule called Sertoli cells

c) The leydig cells which secrete hormones (testosterone)

d) The accessory sex glands pour their secretions over the sperms to form the seminal fluid are: seminal vesicle, prostate ,bulbourethral gland

As regards the Spermatogenesis complete the followings

A. Primary spermatocyte contains 46 Number of chromosomes

B. Spermatids contains 23 Number of chromosomes

C. Process by which the spermatids are transformed from rounded cells to complete sperms called spermiogenesis

D. acrosomal cap is formed by golgi bodies and covers anterior half of head

As regards the Oogenesis complete the followings

a) Primary oocyte is surrounded by a layer of flat follicular cell to form the primordial Follicle

b) The primary oocyte contains …46……number of chromosomes

c) The granulosa cells become separated from the oocyte by a thick hyaline membrane called zona pellucida

d) a sheath of stroma cells formed around the mature follicle and called the theca cells

e) theca interna cells are the main source of estrogen hormone

As regards the OVULATION complete the followings

A. Ovulation takes place ……14……..days before the beginning of the following menstrual cycle.

B. Secondary oocyte together with surrounding zona pellucida and corona radiata is discharged from the ovary

The following factors leading to the rupture of the follicle

1. Increase in size of the follicle.

2. Thinning of the overlying cortex of the ovary.

3. Rise of the pressure of the fluid inside the follicular antrum

There are 2 types of corpus luteum and duration of each

A. corpus luteum of menstruation >>10 days

B. corpus luteum of pregnancy>> 4months

Shortly describe PHASES of the OVARIAN CYCLE

a) follicular phase

b) ovulation

c) luteal phase

Shortly describe PHASES of the MENSTRUAL CYCLE

a) phase of menstrual flow

b) proliferative phase

c) secretory phase

As regards the Fertilization complete the followings

A. It takes place in ampulla of fallopian tube

B. Capacitation means sperm become capable of fertilizing ovum

C. Function of Zona pellucida are

- ❑ **surrounds the oocyte in the ovary**
- ❑ **protects oocyte in the uterine tube**

RESULTS OF FERTILIZATION are

1. The diploid number of chromosomes (46) is restored (23 from the sperm and 23 from the ovum).

2. The sex of the zygote is determined, being either a male (XY) or a female (XX).

3. Cleavage (segmentation) is stimulated to start.

As regards the Blastocyst formation complete the followings

a) The blastocyst formed about the ……6……day of fertilization

b) It having a cavity called blastocele

c) Inner cell mass will develop into embryoblast

d) Outer cell mass differentiated into 2 layers cytotrophoblast and syncytotrophoblast

A patient is being seen due to vaginal bleeding, four months into her pregnancy. High levels of HCG are found but no inner cell mass (embryo) is present. The diagnosis is implantation outside the uterus?

a) -Name given for this condition is ectopic pregnancy

b) -Time of implantation is at the end of 1st week

c) -The commonest site for implantation is upper part of posterior wall

d) Placenta previa means the blastocyst implants very close to the cervix, and as a result the placenta will be placed below the fetus and its four types is centralis., partial, marginalis, low lying

e) Human chorionic gonadotropin (HCG) is secreted by syncytotrophobalst

As regards the chorion complete the followings

⍰ The chorion forms the wall of the chrionic vescicle

⍰ It consists of 3 layers as follows, from outside to inside:

1. syncytiotrophoblast

2. cytotrophoblast

3. somatic layer of extra embryonic mesoderm

As regards the Gastrulation complete the followings

a) Gastrulation means the process by which the bilaminar embryonic disc becomes a trilaminar embryo composed of ectoderm, mesoderm and endoderm (the three primary germ layers).

b) Gastrulation begins with the formation of ……primitive streak……..

c) Time of Gastrulation around day……15…….

Derivatives of ectoderm are

1. Epidermis of the skin, hair and nails.

2. The brain and spinal cord.

3. pia, and arachnoid mater (not the dura).

4. Pituitary gland and medulla of the suprarenal gland (not its cortex)

5. Lining of part of oral cavity and lower 1/2 of anal canal.

6. Epithelium of the nasal cavity.

Derivatives of mesoderm are

1. A11 types of connective tissue, cartilage and bone.

2. All types of muscles.

3.Lymph vessels and lymphatic tissues

4. The whole cardiovascular system: heart, blood vessels and blood cells.

5. Lining of the cavities of the body: pericardium,pleura and peritoneum.

6. Most of the urinary and genital organs and ducts (except parts of the bladder and urethra).7-cortex of the suprarenal gland

DERIVATIVES OF ENDODERM:

1. Most of the lining of the gut (except inlet and outlet).

2. Pancreas and lining of the biliary passages.

3. Lining of the respiratory tract: larynx, trachea and bronchi.

4. Epithelium of the thyroid, parathyroid and thymus glands.

5.bladder and urethra

As regards the neural tube complete the followings

a) Consists of thickening of ectoderm

b) Extends between primitive node and buccopharyngeal membrane

c) Neural tube differentiates into brain and spinal cord

Folding of the disc is due to the following mechanisms

a. Progressive expansion of the amniotic cavity (cavity dorsal to the ectoderm).

b. Rapid growth of the axial area (neural tube and the vertebral column) of the disc more than its periphery.

Results of folding

A)Formation of the gut:

B)Reversal of positions:

C) formation of umblical orifice

Parts of the decidua are

A. capsularis present at superficial end of blastocyst

B. basalis present at deep to blastocyst

C. parietalis present at wall of uterus

Types of villi (According to their structure) are

A. ……1ry……consist of syncytiotrophoblast, cytotrophoblast

B. ……2ry…… consist of syncytiotrophoblast, cytotrophoblast, extra embryonic mesoderm

C. ……tertiary…… consist of syncytiotrophoblast, cytotrophoblast, extra embryonic mesoderm , fetal blood vessels

As regards the placenta complete the followings

A. Foetal part formed by chorion frondosum

B. Maternal part formed by decidua basalis

C. The umbilical cord attached at ……cental…..part

D. Type of blood at Intervillous space is maternal blood

In mid-pregnancy Placental barrier (membrane) formed by

syncytiotrophoblast,

cytotrophoblast,

extra embryonic mesoderm

endothelium fetal blood vessel

Hormone Produced from placenta are

A. ……estrogen…….

B. ……progesteron…….

C. ……hcg……..

D. ……somatommamotropin……..

Structures of the umbilical cord before full-term are

a) ……Wharton jelly…….

b) ……vitellointestinal duct…….

c) ……allantois……..

d) ……umblical vessels……..

e) ……vitelline vessels……

Describe changes occurs to umbilical cord after birth

A. umblical vein>> ligamentum teres

B. umblical arteries>> medial umblical ligaments

C. allantois>> median umblical ligament

As regards the Yolk sac complete the followings

A. Its roof is formed by the endoderm

B. The caudal part of the sac gives a diverticulum called allantois

C. Communicates with the gut through the vitellointestinal duct

D. It's functions includes

1) The primitive gut: derived from the roof of the yolk sac.

2) Allantois: derived from the caudal part of the roof of yolk sac and forms part of the urinary bladder.

3) Primordial sex cells: derived from caudal wall of the yolk sac and migrate to reach the gonads

4) Blood: develops in the mesoderm of the wall of the yolk sac.

As regards the amnion complete the followings

a) Its floor is formed by ectoderm

b) Amniotic fluid is derived from

a -Secretions: by the amniogenic cells of the amnion.

b -Excretions: by the kidneys of the fetus.

c) Functions of the amnion

1) It provides a fluid medium in which the fetus floats freely and grows evenly.

2)It prevents adherence of the fetus to the surrounding uterine walls.

3) It acts as a protective layer against the risk of injury to the fetus from external violence.

4) It maintains the body temperature of the fetus relatively constant.